

Deep Learning for Financial Risk Modeling under Uncertainty: A Comprehensive Survey

Abstract

The integration of deep learning (DL) into financial risk modeling has precipitated a paradigm shift from deterministic point estimates to sophisticated probabilistic frameworks capable of quantifying uncertainty. Traditional econometric models, while interpretable, often struggle with the high dimensionality, non-linearity, and non-stationarity inherent in modern financial markets. This survey provides a comprehensive review of deep learning methodologies designed specifically for financial risk assessment under conditions of aleatoric (data-driven) and epistemic (model-driven) uncertainty. We synthesize evidence from 300 recent academic studies to construct a detailed taxonomy of methodological families, including probabilistic forecasting, robust optimization, deep hedging, generative scenario analysis, and explainable risk attribution. We examine how techniques such as Bayesian neural networks, deep ensembles, evidential regression, and distributional reinforcement learning are being adapted to address specific financial challenges like tail risk estimation, credit default prediction, and derivative pricing. Furthermore, we critically analyze the limitations of current approaches, including computational overhead, calibration issues under regime shifts, and the “black-box” nature of complex architectures. By mapping the landscape of uncertainty-aware deep learning in finance, this paper identifies critical gaps in evaluation standards and deployment safety, offering a roadmap for future research in robust, reliable, and interpretable financial AI.

1. Introduction

Financial risk modeling has historically relied on parametric statistical frameworks, such as Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models and Value-at-Risk (VaR) calculations based on normal or Student’s t-distributions. While these methods provide a foundational understanding of market dynamics, they often fail to capture the complex, non-linear dependencies and heavy-tailed distributions characteristic of financial crises [1], [2]. The advent of deep learning offered a solution to these limitations through universal approximation capabilities, enabling the modeling of intricate patterns in high-dimensional data [3], [4]. However, the initial wave of deep learning applications in finance focused predominantly on maximizing predictive accuracy, often neglecting the quantification of uncertainty—a critical omission in a domain where risk management is paramount [5], [6].

The distinction between aleatoric uncertainty, arising from inherent noise in the data, and epistemic uncertainty, stemming from model ignorance or lack of data, has become central to modern financial AI research [7], [8]. Aleatoric uncertainty is irreducible and must be modeled to provide accurate prediction intervals, particularly for tail risk events [9], [10]. Epistemic uncertainty, conversely, can be reduced with more data or better model specifications and is crucial for detecting out-of-distribution (OOD) scenarios, such as unprecedented market crashes or structural breaks [11], [12]. Recent literature demonstrates that ignoring these uncertainty components can lead to overconfident predictions, catastrophic hedging failures, and regulatory non-compliance [5], [13].

This survey synthesizes the rapidly expanding body of literature on deep learning for financial risk modeling under uncertainty. We categorize the field into several coherent methodological systems: (1) Probabilistic Forecasting and Distributional Regression, which moves beyond point predictions to full density forecasting; (2) Robust Optimization and Decision Making, which incorporates uncertainty directly into objective functions; (3) Deep Hedging and Derivative Pricing, which utilizes

reinforcement learning and neural SDEs for dynamic risk management; (4) Generative Modeling for Scenario Analysis, leveraging GANs and diffusion models for stress testing; and (5) Explainability and Attribution, ensuring that risk models remain interpretable for stakeholders.

The evolution of this field is marked by a transition from simple uncertainty proxies, such as dropout variance, to rigorous Bayesian frameworks, evidential deep learning, and conformal prediction methods [14], [15], [16]. Studies have shown that uncertainty-aware models not only provide better risk estimates but also improve downstream performance in portfolio optimization and trading strategies by dynamically adjusting exposure based on confidence levels [17], [18], [19]. Despite these advances, significant challenges remain, including the computational cost of Bayesian inference, the difficulty of calibrating models under non-stationary regimes, and the lack of standardized benchmarks for uncertainty quantification in finance [7], [20]. This paper aims to provide a rigorous technical overview of these developments, grounded in the evidence provided by 300 distinct academic contributions.

2. Method

The methodological landscape of deep learning for financial risk modeling under uncertainty is diverse, spanning probabilistic architectures, robust optimization frameworks, generative models, and reinforcement learning agents. This section organizes these approaches into coherent families, detailing their technical mechanisms, innovations, and empirical performance as documented in the literature.

2.1 Probabilistic Forecasting and Distributional Regression

The foundation of uncertainty-aware risk modeling lies in the ability to predict full probability distributions rather than single point estimates. This allows for the direct computation of risk metrics such as VaR and Expected Shortfall (ES).

2.1.1 Parametric Distributional Forecasting A primary approach involves parameterizing the output layer of a neural network to predict the parameters of a specific probability distribution. Long Short-Term Memory (LSTM) networks and Transformers are frequently employed to capture temporal dependencies, with output heads mapping to parameters like mean (μ), variance (σ^2), skewness, and kurtosis.

Research indicates that modeling financial returns with skewed Student’s t-distributions significantly outperforms normal assumptions in capturing heavy tails and asymmetry [9]. For instance, LSTM architectures optimized via custom Negative Log-Likelihood (NLL) losses for skewed Student’s t parameters have demonstrated superior Log Predictive Scores compared to CNNs and standard GARCH models on S&P 500 data [9]. Similarly, Mixture Density Networks (MDNs) have been adapted to forecast volatility and returns by modeling the output as a weighted sum of Gaussian components. The “MDNe” framework, which integrates Crossformer encoders with stock code embeddings, effectively approximates dynamic return distributions without rigid shape assumptions, outperforming traditional GARCH variants in Continuous Ranked Probability Score (CRPS) metrics [21].

Extensions to this paradigm include the use of Weibull distributions for Loss Given Default (LGD) modeling in credit risk. By predicting evidential parameters for an Inverse Gamma prior on the Weibull scale, models can derive analytical closed-form solutions for both mean predictions and uncertainty variance, avoiding the need for sampling [22]. Furthermore, the integration of GARCH

dynamics directly into recurrent units, such as in the Integrated GARCH-GRU model, allows for the additive fusion of econometric volatility signals into neural hidden states, capturing volatility clustering while maintaining computational efficiency [23].

2.1.2 Non-Parametric and Quantile-Based Approaches When distributional assumptions are too restrictive, non-parametric methods such as quantile regression offer a flexible alternative. Deep quantile regression networks optimize pinball loss functions to estimate specific conditional quantiles, which can then be aggregated to form empirical cumulative distribution functions (CDFs).

The “ReSGA” (Retrieval-Enhanced Self-Grouping Autoencoder) model exemplifies this approach by employing a unified framework that enforces joint VaR and ES forecasting through Fissler-Ziegel loss functions. By utilizing a retrieval mechanism to access long-term spatial-temporal dependencies across asset groups, ReSGA achieves state-of-the-art performance in tail risk forecasting, scaling primarily with data complexity rather than model size [24]. Similarly, the “FutureQuant Transformer” adapts attention mechanisms to Limit Order Book data, shifting from point prediction to distributional forecasting via quantile regression, thereby providing superior risk assessment for futures markets [25].

Advanced techniques combine quantile regression with Extreme Value Theory (EVT). The Extreme Quantile Regression Network (EQRN) employs a two-stage framework: first estimating intermediate quantiles via recurrent networks, and second, modeling the conditional Generalized Pareto Distribution (GPD) parameters for the tails. This hybrid approach enables accurate extrapolation for extreme conditional quantiles, outperforming standard gradient boosting and linear models in modeling sequential tail risks [26]. Additionally, Deep Huber Quantile Regression Networks (DHQRN) utilize generalized Huber scoring functions to nest quantile and expectile regression, yielding elicitable predictions that generalize well across diverse datasets [27].

2.1.3 Evidential Deep Learning and Higher-Order Uncertainty Evidential Deep Learning (EDL) represents a significant advancement by treating network outputs as parameters of a higher-order evidence distribution (e.g., Dirichlet for classification, Normal-Inverse-Gamma for regression). This allows for the explicit decomposition of uncertainty into aleatoric and epistemic components within a single forward pass.

The “ProbFM” framework integrates EDL with Normal-Inverse-Gamma priors into a Transformer backbone for time-series forecasting. By employing adaptive evidence accumulation and coverage loss, ProbFM achieves explicit uncertainty decomposition without sampling, resulting in improved Sharpe and Sortino ratios on cryptocurrency data compared to MSE baselines [14]. In credit risk, “riskEDL” replaces softmax outputs with evidence vectors parameterizing Dirichlet distributions, introducing dual-head architectures to predict sample-specific pignistic priors. This approach has been shown to reduce misclassification costs by over 55% compared to cost-sensitive training, effectively bridging uncertainty estimation with risk minimization [15].

However, the validity of EDL methods has been scrutinized. Recent studies suggest that unregularized inner loss minimization in EDL can suffer from non-identifiability, where multiple parameter sets yield identical predictive distributions, potentially leading to miscalibrated epistemic uncertainty [28]. To address this, frameworks like “Credal Bayesian Deep Learning” (CBDL) construct finitely generated credal sets via convex hulls of extreme elements, enabling robust uncertainty quantification through imprecise probability theory [29].

2.2 Robust Optimization and Decision Making under Uncertainty

Incorporating uncertainty into the decision-making process is as critical as estimating it. Robust optimization frameworks modify standard objective functions to penalize model misspecification or worst-case scenarios.

2.2.1 Uncertainty-Aware Portfolio Optimization Traditional mean-variance optimization is highly sensitive to estimation errors in expected returns and covariance matrices. Deep learning approaches address this by integrating uncertainty estimates directly into the portfolio construction process.

The “Uncertainty-Aware Strategy with Subsampling” framework superimposes outer uncertainty measures, such as Entropic Risk or Conditional Value-at-Risk (CVaR), on the space of candidate models. By approximating model distributions via subsampling analogous to bootstrapping, this method modifies standard objective functions to enhance robustness against model misspecification, outperforming plug-in and mixture strategies in pair trading utilities [30]. Similarly, “Uncertainty-Adjusted Sorting” constructs asset-specific prediction intervals and uses upper/lower bounds for cross-sectional ranking instead of point predictions. This approach reduces portfolio volatility more than it reduces mean returns, thereby increasing the Sharpe ratio in US equity panels [19].

Deep reinforcement learning (DRL) has also been adapted for robust portfolio optimization. The “Diffusion-Augmented Reinforcement Learning” (DARL) framework employs conditional Denoising Diffusion Probabilistic Models (DDPM) to generate synthetic stress scenarios parameterized by crash intensity. These scenarios augment the replay buffer for a Proximal Policy Optimization (PPO) agent, enabling the learning of policies that are resilient to tail risks and data scarcity [31]. Furthermore, “Risk-Aware DRL with PPO” integrates Hidden Markov Models (HMM) to model hidden market states within the DRL loop, directly optimizing the Sharpe ratio while stabilizing policy updates in volatile environments [32].

2.2.2 Distributionally Robust Optimization (DRO) Distributionally Robust Optimization seeks strategies that perform well under the worst-case distribution within an ambiguity set, often defined by Wasserstein distances.

In the context of statistical arbitrage, “P-robust G-arbitrage” constructs data-driven ambiguity sets using Wasserstein balls around empirical measures. Deep neural networks approximate the conditional super-replication problem, enabling scalable optimization of trading strategies under model ambiguity [33]. For systemic risk, “Robust Mean-Field Control” formulates discrete-time control as a lifted robust Markov Decision Process on the space of probability measures, constructing uncertainty sets for common noise laws using Wasserstein perturbations [34].

Theoretical advancements include the formulation of “Lambda Rényi Entropic Value-at-Risk” (Λ -EVaR), which unifies flexible confidence levels with higher-moment sensitivity via Rényi entropy. This measure provides closed-form worst-case bounds under Wasserstein uncertainty sets, extending beyond standard EVaR formulations [35]. Additionally, “Robust Generalized Quantiles” transform classical quantiles and expectiles into robust counterparts by incorporating Wasserstein distance penalization, maintaining convexity and coherence properties essential for risk management [36].

2.2.3 Risk Budgeting and Dynamic Allocation Dynamic risk budgeting strategies allocate capital based on time-varying risk contributions. “Dynamic Risk Budgeting via Actor-Critic Deep

Learning” defines dynamic risk contributions using one-sided Gâteaux derivatives of coherent distortion dynamic risk measures. By recasting dynamic risk budgeting as a sequence of convex optimization problems solved by actor-critic networks, this method ensures self-financing strategies that converge to target risk budgets [37].

End-to-end learning frameworks further integrate prediction and optimization. “End-to-End Risk Budgeting” (E2E-RB) embeds risk-budgeting convex optimization as a differentiable implicit layer within a neural network. This joint training minimizes task-specific loss rather than prediction error, outperforming nominal risk parity strategies in real market ETF datasets [38]. Similarly, “Financially Guided Deep Portfolio Optimization” integrates differentiable surrogates for Sharpe/Omega ratios and CVaR into loss functions, employing walk-forward validation to ensure robust out-of-sample performance [39].

2.3 Deep Hedging and Derivative Pricing

Deep learning has revolutionized derivative pricing and hedging, particularly in incomplete markets where traditional no-arbitrage arguments are insufficient.

2.3.1 Deep Hedging with Risk Measures Deep hedging utilizes neural networks to learn optimal hedging strategies by minimizing a risk measure of the hedging error. “Deep Equal Risk Pricing” (Deep ERP) deploys dual LSTM networks to approximate optimal trading policies for long and short positions separately, minimizing residual hedging risk quantified by convex risk measures like CVaR [40]. This approach has been extended to handle multiple hedging instruments, showing that hedging long-term puts with short-term options significantly reduces equal risk prices under jump risk [40].

To address tail risk specifically, “Deep Tail-Risk Hedging via CVaR Minimization” formulates hedging as a stochastic optimization problem minimizing expected 1-Day 99% CVaR. By employing block-bootstrapped market paths to preserve fat tails, this method generates synthetic crisis scenarios that improve hedging performance during market crashes compared to Gaussian simulations [41]. “Tail-Safe” hedging introduces a white-box Control Barrier Function (CBF) safety layer within a distributional RL framework, enforcing state-wise safety constraints while optimizing CVaR [42].

2.3.2 Neural SDEs and Physics-Informed Pricing Neural Stochastic Differential Equations (SDEs) provide a flexible framework for modeling asset dynamics. “Bayesian Neural SDE Calibration” defines drift and diffusion coefficients as neural networks, constructing posterior distributions over weights to provide robust price bounds through mixture distributions [43]. This Bayesian formulation transforms calibration into posterior inference, offering tighter implied volatility bounds compared to classical methods.

For path-dependent options, “Non-linear Generalized Affine (NGA)” frameworks formulate robust pricing as the supremum of expectations over a class of semimartingale laws. By deriving functional non-linear Kolmogorov equations via Functional Itô Calculus, these models solve fully non-linear path-dependent PDEs using deep backward SDEs, matching finite difference schemes with visible efficiency gains [44], [45]. “Physics-Informed Acoustic Models” (PIAM) further integrate physical constraints, such as the Westervelt equation, as inductive biases to penalize latent space trajectories inconsistent with nonlinear acoustic dynamics, enhancing market interpretability [46].

2.3.3 Uncertainty-Aware Hedging Strategies Recent work emphasizes the role of uncertainty in executing hedging decisions. “Uncertainty-Aware Deep Hedging” trains ensembles of LSTM networks and uses per-step ensemble standard deviation as an uncertainty proxy to construct blended hedge ratios. This CVaR-driven adaptive blending mechanism switches between aggressive and conservative hedging based on model agreement, statistically significantly improving CVaR over Black-Scholes baselines [47]. Similarly, “Adaptive Nesterov Accelerated Distributional Deep Hedging” (ANADDH) integrates distributional RL with adaptive Nesterov acceleration, modeling the full reward distribution to capture tail risks better than expected value maximization [48].

2.4 Generative Modeling for Scenario Analysis and Stress Testing

Generative models are increasingly used to simulate realistic market scenarios for stress testing and risk assessment, overcoming the limitations of historical simulation.

2.4.1 GANs and VAEs for Financial Synthesis Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are employed to generate synthetic financial time series that preserve statistical properties such as volatility clustering and heavy tails. “TimeGAN” has been shown to outperform VAEs and ARIMA-GARCH in replicating distributional shapes and auto-correlation structures of S&P 500 log-returns, enabling robust downstream portfolio optimization [49].

For economic scenario generation, custom GAN architectures replace parametric models, automatically learning complex dependency structures from empirical data without explicit copula selection [50]. “PredACGAN” reverses standard ACGAN structures, using historical return sequences as conditional labels to generate predictive distributions, enabling entropy-based risk filtering that significantly improves yearly returns [51]. However, limitations exist; plain historical simulation often outperforms deep generative models in distribution fidelity due to noise in neural network training, highlighting the need for careful architectural design [52].

2.4.2 Diffusion Models and Counterfactuals Diffusion models offer high-fidelity generation capabilities. “LLM-Generated Counterfactual Stress Scenarios” utilize a hybrid Prompt-RAG pipeline where Large Language Models generate structured macro shocks subject to plausibility gates. This approach separates volatility, linear factor, and nonlinear text-driven effects, providing narrative-stabilized tail risk assessments [53].

In the context of reverse stress testing, “Vine Copula Reverse Stress Testing” models joint distributions of risk factors and portfolio loss using R-vine copulas. By defining stress scenarios as conditional modes, this method enables flexible high-dimensional dependence modeling beyond elliptical constraints, outperforming Gaussian kernel methods in bias and variance [54]. “Generative Adversarial Regression” (GAR) further casts conditional risk estimation as a regression problem under strictly consistent scores, enabling direct optimization of risk functionals like VaR and ES [55].

2.5 Systemic Risk and Network Analysis

Modeling systemic risk requires capturing the interdependencies between financial institutions. Deep learning facilitates the analysis of complex financial networks.

2.5.1 Graph Neural Networks for Contagion Graph Neural Networks (GNNs) are pivotal in modeling interbank networks and contagion effects. “Conditional DGNN for Variation Margin Forecasting” integrates GC-LSTM modules to encode network topology and node features, enabling accurate multi-step margin call forecasting under specific stress scenarios by conditioning on future reference rate trajectories [56]. “Robust Graph Neural Networks” employ multi-head attention to weight high-risk neighbor influence, effectively revealing hidden risk clusters in synthetic economic networks [57].

For systemic risk measurement, “Deep Learning Primal-Dual Algorithms” approximate optimal risk allocations and Radon-Nikodym derivatives using neural networks, solving multivariate utility optimization problems without imposing theoretical measurability constraints a priori [58], [59]. “Transformer-based CoVaR” integrates raw text embeddings from LLMs with numerical bank returns, utilizing self-attention to capture nonlinear dependencies between textual context and financial variables, showing significant deviations from return-only models during crises [60].

2.5.2 Topological and Interaction Uncertainty Beyond standard network metrics, topological data analysis offers new insights. “Topological VaR Distance” (TVaRD) constructs Vietoris-Rips complexes from normalized returns to compute persistent homology diagrams, detecting multi-scale market stress invisible to statistical moments [61]. “Interaction uncertainty in financial networks” models banks via stochastic dynamics with integral interactions, quantifying how unknown interaction matrices act as additional noise sources increasing systemic stress [62].

2.6 Explainability and Risk Attribution

The “black-box” nature of deep learning poses challenges for regulatory compliance and trust. Explainable AI (XAI) techniques are being adapted to provide risk attributions.

2.6.1 Axiomatic Risk Attribution “Baseline Shapley for Risk Attribution” (BShap-RAM) extends Shapley values to map feature subsets to risk measures rather than mean predictions. This formalization demonstrates that risk attribution requires redefining characteristic functions, as the linearity axiom becomes incompatible with risk measures like CVaR [63]. “Gradient-based Uncertainty Attribution” (UA-Backprop) decomposes total uncertainty into epistemic and aleatoric components via backpropagation, satisfying completeness while avoiding vanishing gradients [64].

2.6.2 Multimodal and Temporal Explainability “Uni-FinLLM” utilizes a unified multimodal encoder to align heterogeneous data (price, text, macros) into a shared embedding space, enabling transfer of patterns between asset-level behaviors and systemic stress factors while maintaining interpretability through modular task heads [65]. “A Time Series Approach to Explainability” proposes X-functions computing time-dependent partial derivatives, aggregating sensitivities across ensembles to reveal underlying linear forecast heuristics obscured by initialization noise [66].

2.7 Calibration, Evaluation, and Theoretical Foundations

Ensuring that uncertainty estimates are well-calibrated is critical. Theoretical works provide bounds and convergence guarantees for these methods.

2.7.1 Calibration and Scaling Laws “Scaling Laws for Uncertainty” demonstrate that predictive uncertainties follow robust power-law trends with dataset size across diverse architectures,

though epistemic uncertainty decay rates vary by inference method [20]. “On the Calibration of Epistemic Uncertainty” identifies that posterior approximation errors in SGD cause uncertainty collapse, proposing conflictual deep ensembles to restore monotonicity principles [67]. “Risk-Controlled Early Exiting” formalizes early-exit networks as threshold-based predictors subject to statistical risk bounds, providing distribution-free guarantees for efficiency [68].

2.7.2 Theoretical Risk Bounds Theoretical analyses establish convergence rates for deep risk models. “Super-fast Rates of Convergence” prove that under hard-margin conditions, DNNs can achieve arbitrarily fast polynomial convergence rates for excess risk [69]. “Minimax Optimal High-Dimensional Classification” derives minimax excess misclassification risk rates for sparse DNNs, showing dependence on intrinsic dimension rather than ambient dimension [70]. “Risk Bounds For Distributional Regression” provides uniform convergence rates for neural network-based distributional regression under structural constraints like monotonicity [71].

3. Challenges and Outlook

Despite significant progress, the field of deep learning for financial risk modeling under uncertainty faces several persistent challenges and open problems.

3.1 Computational Complexity and Scalability

A primary bottleneck is the computational cost of rigorous uncertainty quantification. Bayesian Neural Networks (BNNs) and Deep Ensembles, while effective, incur substantial training and inference overheads [7], [8]. Methods like MCMC sampling for Bayesian calibration are often too slow for real-time risk monitoring or high-frequency trading applications [72], [73]. While approximations like Monte Carlo Dropout and Variational Inference reduce costs, they often sacrifice calibration quality or fail to capture complex posterior geometries [74], [75]. Future research must focus on developing scalable approximations, such as the “Hyper-Diffusion Models” which approximate deep ensembles via Bayesian hyper-networks with minimal training overhead [76], or efficient propagation methods like “Factor Graph Uncertainty Propagation” [77].

3.2 Non-Stationarity and Regime Shifts

Financial markets are inherently non-stationary, with structural breaks rendering historical data less relevant for future prediction. Many deep learning models assume i.i.d. data or stationary distributions, leading to performance degradation during crises or regime shifts [78], [79]. “Conditional Adversarial Fragility” studies show that adversarial vulnerability amplifies systematically under macroeconomic stress, even when baseline performance remains stable [78]. Addressing this requires models that can adapt online or detect regime changes dynamically. Approaches like “Risk-Sensitive Specialist Routing,” which gates models based on volatility regimes, offer a promising direction [79], as do meta-learning frameworks that adapt to unknown market regimes via soft clustering [80].

3.3 Calibration and Reliability under Tail Events

Accurately quantifying uncertainty in the tails of the distribution remains elusive. Many models are well-calibrated in the body of the distribution but fail to provide reliable confidence intervals for extreme events [81], [82]. “Evaluating financial tail risk forecasts” highlights that standard

tests like Diebold-Mariano have negligible power against models underestimating tail risk at extreme quantiles with short evaluation windows [81]. Furthermore, “Accuracy estimation of neural networks by extreme value theory” reveals that classical estimators often violate the finite upper bound constraints of error distributions, necessitating constrained maximum-likelihood estimators [82]. Developing evaluation metrics and loss functions specifically tailored for tail calibration, such as those based on elicitable functionals [55] or extreme value theory [26], is a critical area for future work.

3.4 Interpretability and Regulatory Compliance

The “black-box” nature of deep learning hinders its adoption in regulated environments where model interpretability is mandatory (e.g., Basel III, SR 11-7). While XAI techniques like Shapley values and attention maps provide insights, they often lack stability or fail to capture the non-linear interactions inherent in risk models [3], [63]. “Black-box model risk in finance” warns that post-hoc explanations can provide false security and that neural networks lack the simulatability and decomposability of parametric models [5]. Future research must prioritize inherently interpretable architectures or develop rigorous certification frameworks, such as “Risk-Averse Certification of Bayesian Neural Networks,” which integrates CVaR into formal verification [13].

3.5 Data Scarcity and Synthetic Data Quality

High-quality labeled data for rare events (e.g., defaults, crashes) is scarce. While generative models offer a solution, their ability to faithfully reproduce tail dependencies and extreme correlations is not guaranteed. “Deep generative modeling for financial time series” notes that plain historical simulation often outperforms deep learning models in distribution fidelity due to noise and instability in neural generation [52]. Ensuring that synthetic data preserves the “stylized facts” of financial markets, particularly in the tails, remains an open challenge. Hybrid approaches combining physics-informed constraints with generative modeling, such as “Physics-Informed Acoustic Models” [46] or “Neural SDEs” [43], may offer a path forward.

3.6 Integration of Multimodal and Alternative Data

The increasing availability of unstructured data (news, satellite imagery, earnings call audio) presents opportunities for richer risk models. However, effectively fusing these modalities while quantifying the uncertainty associated with each source is complex. “Uni-FinLLM” and “RiskLabs” demonstrate the potential of multimodal LLMs, but challenges remain in handling noise, misinformation, and the varying reliability of different data streams [65], [83]. Future frameworks must develop robust fusion mechanisms that can weigh information sources dynamically based on their estimated uncertainty.

4. Conclusion

The integration of deep learning into financial risk modeling has evolved from a focus on predictive accuracy to a sophisticated engagement with uncertainty. This survey has synthesized a vast array of methodologies, demonstrating that uncertainty-aware deep learning is not merely an add-on but a fundamental requirement for robust financial decision-making. From probabilistic forecasting with skewed distributions and evidential regression to robust optimization via Wasserstein ambiguity sets and deep hedging with CVaR objectives, the field has developed a rich toolkit for addressing aleatoric and epistemic risks.

Key innovations include the decomposition of uncertainty into actionable components, the use of generative models for stress testing, and the application of graph neural networks for systemic risk analysis. However, significant hurdles remain. Computational efficiency, calibration under non-stationarity, tail risk reliability, and interpretability are critical areas requiring further investigation. The future of financial risk modeling lies in the development of hybrid systems that combine the flexibility of deep learning with the rigor of economic theory and the robustness of Bayesian inference. As the field matures, standardized benchmarks for uncertainty quantification and rigorous regulatory frameworks will be essential to ensure the safe and effective deployment of these powerful technologies in the global financial system.

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